

Violations for Continuous Candidates in Constraint-based Phonology

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Abstract

Constraint-based phonology normally assigns violations to discrete candidates. At the phonetics-phonology interface, however, many observables are continuous, so pre-evaluative discretization can impose arbitrary boundaries and false local homogeneity. This article argues that continuous candidate spaces therefore require an explicit theory of violation computation. I propose a model in which candidate availability is represented by an empirical density over acoustic space, perceptual and articulatory constraints are defined as continuous functions, and violations are derived by linking constraint values to empirical density. Among several transparent link functions, an exponentialized relation provides the most coherent violation object under the present criteria. A two-formant implementation over Brazilian Portuguese pretonic /e/ shows that weighted harmonic and MaxEnt-style evaluation can operate directly over a continuous vowel category, yielding region-sensitive well-formedness and a fitted output density without pre-evaluative binning while preserving category-internal gradient structure in phonological evaluation itself.

Keywords: continuous-candidate phonology; phonetics-phonology interface; constraint-based phonology; empirical density; computational phonology; Harmonic Grammar; Maximum Entropy phonology; kernel density estimation

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1 Introduction

Constraint-based phonology normally evaluates discrete candidates. In Optimality Theory, Harmonic Grammar and Maximum Entropy approaches, candidates are compared by constraints, violations are ranked or weighted, and the grammar selects or probabilistically favors some outputs over others (Prince and Smolensky, 1993; Kager, 2004; Flemming, 2021; B. Hayes and Wilson, 2008; Magri, 2018). That architecture works smoothly when the objects of evaluation come already delimited as a finite set. At the phonetics–phonology interface, however, many relevant observables are continuous. Vowel formants, duration, pitch, timing and intensity occupy continua rather than arriving as bounded rows in a tableau. Once candidates remain continuous, the grammar can no longer inherit violations from a pre-enumerated candidate list. The theoretical problem addressed here is therefore not whether continuous-candidate phonology is possible in principle, but what a violation is when the candidate space is continuous.

A common response is to discretize the continuum before evaluation, treating intervals as candidates in an otherwise familiar grammar (Boersma, P. Escudero, and R. Hayes, 2003). That move is understandable, but it is not innocent. Pre-evaluative binning can force phonetically adjacent values into different candidates while treating more distant values as equivalent simply because they fall inside the same interval. The result is not only loss of detail. It is the imposition of false local homogeneity and boundary artifacts on a space whose structure may vary continuously. If weighted phonological evaluation is to operate directly over continuous candidate structure, violations must be defined without relying on bins to supply them in advance.

The proposal developed here supplies that missing object. Candidate availability is represented as an empirical probability density over the relevant phonetic space, estimated from observed data. Perceptual and articulatory pressures are formulated as continuous constraint functions over that same space. Violations are then defined by an explicit link between continuous constraint value and empirical density. The resulting quantity can feed ordinary weighted phonological machinery: pointwise violations can be integrated over continuous domains, combined into harmonic scores and transformed into a continuous MaxEnt output density. The paper’s central

claim is therefore narrow but consequential: constraint-based phonology lacks an explicit violation theory for continuous candidate spaces, and such a theory can be supplied without abandoning the familiar architecture of weighted evaluation.

This use of empirical density creates an obvious resemblance to exemplar-like approaches, since both take observed token structure seriously. The resemblance should be acknowledged, but the division of labor is different. In an exemplar account, the token cloud may itself function as representational content for the category. In the present framework, by contrast, empirical density serves as an availability field: it indicates where candidate mass is concentrated so that independently specified grammatical pressures can be evaluated against that concentration. The formal object of interest is not the density alone, but the violation quantity produced by its interaction with continuous constraints. The contribution is thus an explicit account of the grammar–data interface in continuous space rather than a re-description of category structure in exemplar terms.

This point also clarifies why the proposal is phonological rather than merely statistical. The paper does not replace weighted-constraint architecture with kernel density estimation plus arbitrary geometric functions. Constraints remain the source of grammatical pressure; violations remain the quantities that weighted evaluation operates over; harmony remains a weighted sum; and MaxEnt-style modeling remains a probabilistic transform of weighted penalties. What changes is that the violation object can no longer be presupposed once candidates cease to be discrete. The proposal therefore leaves the downstream logic of weighted phonology intact while making explicit a formal precondition that discrete tableaux normally hide. The contribution of this article is therefore to the formal architecture of constraint-based phonology at the phonetics–phonology interface: it specifies what counts as a violation in continuous candidate space, rather than merely introducing a new numerical modeling routine.

The article also positions itself with respect to earlier work on continuous phonetic spaces. Prior research has shown that phonological analysis can extend over continuous domains and that constraints can be formulated as explicit functions over those domains, including Flemming (2001) and Lefkowitz (2017). The question pursued here is narrower. Once continuous constraints

are available and candidate structure is represented by an empirical distribution rather than a pre-delimited list, what local quantity should count as a violation? In that sense, the paper is complementary to learning-oriented and stochastic traditions within constraint-based phonology (Boersma, 1998; Boersma, P. Escudero, and R. Hayes, 2003; B. Hayes and Wilson, 2008). Those frameworks address how constraint systems are ranked, weighted or learned. The present article addresses the logically prior question of what local violation magnitude exists to be weighted or learned in the continuous case.

The empirical implementation uses attested Brazilian Portuguese pretonic /e/ data in a two-formant space. F1 and F2 are used not because the proposal is restricted to formants, but because they provide a familiar continuous domain in which the formal problem can be shown transparently over real acoustic observations. The empirical case is therefore an implementation domain rather than the paper's primary argumentative burden. Its role is to show that once violations are defined explicitly, weighted phonological evaluation can operate directly over a continuous vowel category rather than over a discretized surrogate.

The argument proceeds by comparing a small family of transparent link functions between continuous constraint value and empirical density. A usable violation quantity should remain non-negative, attenuate in empirically dense regions, remain stable in sparse or extrapolative regions, support later weighted evaluation and remain interpretable as a magnitude of violation. Under those desiderata and in the present implementation, the exponentialized relation emerges as the strongest admissible option. That result should be understood as a solution to a specific methodological problem: how to define violations once candidates remain continuous.

The rest of the paper develops that solution in sequence. Section 2 sharpens the discretization problem and shows why pre-evaluative binning distorts local evaluation. Section 3 constructs the empirical candidate space from observed acoustic data. Section 4 introduces continuous perceptual and articulatory constraints, and Sections 5–6 derive a usable violation object by linking those constraints to empirical density. Sections 7–8 show how the resulting violations feed ordinary harmonic and MaxEnt-style evaluation, and Section 9 illustrates the resulting region-sensitive

interpretation over the present /e/ space. Section 10 limits the scope of the empirical claim, and Section 11 returns to the broader theoretical payoff: continuous candidate structure can be interpreted phonologically once violation computation is made explicit.

2 Continuous candidates and discretization

Constraint-based phonology is usually implemented over discrete candidates. In Optimality Theory, candidates are compared under a ranking of constraints; in Harmonic Grammar and Maximum Entropy approaches, they are evaluated by weighted violation profiles (Prince and Smolensky, 1993; Kager, 2004; Flemming, 2021; B. Hayes and Wilson, 2008). In that family of models, evaluation normally presupposes that the relevant candidate set is already listed or otherwise delimited in advance. That architecture works smoothly when the objects of evaluation are discrete. At the phonetics–phonology interface, however, many of the relevant dimensions are continuous: vowel formants, duration, pitch, intensity, and timing are distributed over continua rather than given as a finite inventory of pre-delimited outputs.

A standard response is to discretize that space into bins or other discrete acoustic candidates and then treat those discrete regions or values as candidates (Boersma, P. Escudero, and R. Hayes, 2003; Boersma, 1998; Boersma, 2005; Lefkowitz, 2017; Braver, 2019; Zhang, 2007; Flemming, 2001; P. R. Escudero, 2005; Shchupak, 2022). The convenience of that move is obvious, but its cost is methodological. The problem is not discretization in the abstract; it is the homogeneous treatment imposed inside each bin and the stepwise discontinuities imposed at bin boundaries. Although the examples and toy illustrations below uses formant values in Hz, and the content of this paper itself will narrow its scope to this specific phonetic detail, the objection is not specific to vowel formants. The same methodological problem arises whenever naturally continuous phonetic dimensions are converted into pre-evaluative bins or other discrete candidates, whether along duration, pitch, timing, intensity, or higher-dimensional acoustic combinations.

A well-known version of this logic appears in acquisition-oriented work that maps continuous

auditory input onto a smaller set of perceived categories (Boersma, P. Escudero, and R. Hayes, 2003). Figure 1 reproduces the basic schematic from Boersma, P. Escudero, and R. Hayes (2003). It is useful here because it makes the relevant methodological move visible: once a categorical partition is imposed on a continuous space, the grammar inherits the edges of that partition.

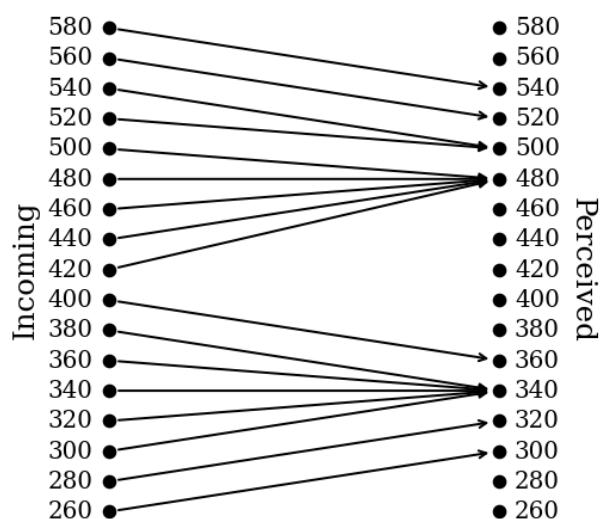


Figure 1: Warping of continuous incoming inputs into perceived categories, adapted from Boersma, P. Escudero, and R. Hayes (2003).

The resulting artifact can be illustrated with a toy F1 interval from 300 to 500 Hz, discretized into two candidates, 300–400 Hz and 401–500 Hz. If one toy constraint penalizes 350 Hz and another penalizes 450 Hz, the two bins receive different violation profiles simply because they have been turned into bounded candidates.

Table 1: Toy discrete Optimality Theory tableau for a discretized acoustic interval.

<i>/input/</i>	*350 Hz	*450 Hz
a. 300–400 Hz	*!	
☞ b. 401–500 Hz		*

The artifact appears at once: 400 Hz and 401 Hz become categorically different candidates even though they are adjacent in the original continuum, while 350 Hz and 399 Hz may be treated

alike merely because they fall inside the same bin. The same difficulty becomes sharper under weighted probabilistic evaluation, such as MaxEnt and Harmonic Grammar.

Table 2: Toy weighted probabilistic tableau illustrating a boundary artifact after discretization.

<i>/input/</i>	*350 Hz (w=1)	*450 Hz (w=3)	H	P
☞ a. 300–400 Hz	-1		-1	88.1%
b. 401–500 Hz		-3	-3	11.9%

Here a value at 400 Hz inherits one probability and a value at 401 Hz another, despite the negligible phonetic distance between them. One could argue, for instance, that we could employ smaller bins – and the smaller they get, the more granularly we could distribute violations and, consequently, probabilities for each of these candidates of phonetic production. But at which point would the increasingly granular phonetic candidates simply become analytically unsustainable, bordering on infinity? A continuous method is therefore conceptually cleaner. It preserves the continuity of the physical and mechanical reality of sound, and of local evaluation where discretized methods force stepwise discontinuities. The consequence of discretization, therefore, is not only loss of phonetic detail, but loss of any ready-made violation object once evaluation is allowed to proceed over a continuous candidate domain without pre-binning.

The objection should be stated carefully. It is not a denial that perception in the world may involve relatively abrupt category transitions (Stevens, 1972). The point is narrower: pre-evaluative binning adds an additional homogeneity assumption. All values inside a bin are treated as though they incur the same violations simply because they were grouped together before evaluation. That is a modeling decision, not a fact about the phenomenon itself. A continuous approach preserves the possibility that neighboring values differ only slightly in evaluation while still allowing later summaries over intervals when those are analytically useful.

Seen from that perspective, discretization does not merely coarse-grain the space. It can also falsely homogenize local structure. Once an interval is promoted to candidate status, internally distinct portions of that interval inherit a single violation profile even if the relevant grammatical

pressure should vary continuously across them. Values that are locally unlike may therefore be treated as equivalent simply because they were placed in the same pre-delimited region, while values that are locally almost identical may be assigned categorically different profiles because they straddle an analyst-imposed edge. The methodological cost is therefore not exhausted by “loss of information”; it lies in replacing a continuous local geometry with analyst-imposed uniformity.

This matters because the paper’s central question is about violations, not just representation. In a discretized model, the candidate partition silently supplies the units to which violations are assigned. Once that partition is removed, violations can no longer be inherited from pre-delimited candidates. They must be defined locally over the space itself. A continuous model therefore requires more than finer resolution: it requires an explicit account of how grammatical pressure is computed over a continuous candidate domain.

The next section turns to the empirical side of that problem. If candidates are not pre-segmented into bins, then the observed data must themselves be represented continuously. In the present proposal, empirical density provides that representation of candidate availability over the acoustic space. The next question can then be stated precisely: how should a continuous constraint and an empirical distribution interact so that violations remain interpretable, non-negative, and usable in later weighted evaluation?

3 Empirical candidate space

Once candidates are no longer inherited from pre-binned intervals, the model requires an empirical object that can stand in for the candidate space itself. In the present framework, that object is an estimated probability density over the relevant acoustic dimensions. The point is not to replace grammar with distributional description. It is to supply a continuous availability field over which independently specified constraints can later operate. The present corpus serves as an empirical implementation of that idea over attested acoustic data, not as a language-wide phonological survey.

The current implementation is two-dimensional, using F1 and F2. This choice is both

practical and theoretically transparent: the first two formants provide a familiar vowel space and are sufficient to show how continuous evaluation can operate over more than one phonetic dimension at once. Nothing in the proposal depends in principle on F1 and F2 alone. The same architecture could be extended to other continuous dimensions, including duration, intensity, pitch, timing, or higher-dimensional combinations of measurements.

Table 3: Current implementation settings for the empirical candidate space.

Corpus size	114 tokens
Vowel	/e/
Speakers	three adult male speakers (IDs 1, 3, 5)
Acoustic domain	F1 = 250–543 Hz; F2 = 908–1987 Hz
Preprocessing	z-score standardization
KDE bandwidth	0.2

The empirical material consists of Brazilian Portuguese pretonic /e/ tokens. This vowel is a useful implementation domain because its acoustic realization is gradient rather than point-like: pretonic /e/ participates in contextual raising and related variation, yielding continuous spread across the relevant portion of the vowel space (Madruga, 2017). That makes it suitable for demonstrating the formal problem addressed here. A larger corpus would reshape the estimated surface, but it would not change the theoretical issue: once candidates remain continuous, candidate availability must itself be represented continuously.

Before density can be estimated, the observations must be brought onto a common scale. In the present implementation, that preprocessing choice is made in the service of stable joint estimation rather than as a claim about the uniquely correct normalization of acoustic space. Lobanov normalization was initially tested, since it is standard in acoustic phonetics for reducing speaker-dependent differences in vowel spaces (Lobanov, 1971). In the present sample, however, the intra-speaker Lobanov-transformed values yielded a singular covariance matrix at the kernel-density stage, so the joint density could not be estimated stably in that form. Z-score standardization was

therefore adopted because it preserved workable variability while placing the dimensions on a common scale.

That choice matters for the formal architecture in a limited but important sense. Standardization ensures that the density estimate and the later constraint functions are evaluated in the same coordinate system: all formant values, including the constraint parameters introduced below, are transformed into z-scored coordinates before computation, so that $C_i(x)$ and $\hat{f}(x)$ are defined over the same space throughout. Standardization does not itself solve the problem of how density values and constraint values become jointly usable. That task belongs to the definition of violation. What preprocessing supplies is a stable and interpretable domain in which that later interaction can be computed.

Once standardized, the observations are used to estimate a continuous probability density function by kernel density estimation (KDE), a non-parametric procedure that constructs a smooth distribution from local kernel contributions around observed points (Hastie, Tibshirani, and Friedman, 2008). In one dimension, the estimator may be written as

$$\hat{f}_X(x_0) = \frac{1}{N\lambda} \sum_{i=1}^N K_\lambda(x_0, x_i),$$

where $x_1, \dots, x_N$ are the observed samples, K_λ is a kernel centered on each observation and λ is the bandwidth parameter. Intuitively, each token contributes local density around its position, and the estimated distribution is the sum of those local contributions. Larger values of λ produce a smoother curve with less local detail, whereas smaller values preserve finer structure at the cost of greater sensitivity to noise (Hastie, Tibshirani, and Friedman, 2008). In the present implementation, the bandwidth is fixed at 0.2.

For the current dataset, the relevant empirical object is the joint density over F1 and F2, denoted $\hat{f}_{\text{data}}(F1, F2)$. This density does more than summarize the observations descriptively. It represents where candidate mass is empirically concentrated and therefore where there are candidates worth evaluating. Dense regions correspond to portions of the space in which the

sample accumulates; sparse regions correspond to areas that are weakly represented or unattested in the corpus. Figure 2 shows the resulting empirical distribution. The density is estimated over standardized values but plotted back on the original acoustic scale for interpretability; for visual clarity, the two formants are displayed separately even though the estimate itself is joint.

This is the role of density in the present framework. It is not itself a constraint, not the grammar and not a substitute for phonological structure. Without it, a continuous constraint would assign values over an abstract space without any indication of where empirical candidate mass lies. With it, the model can distinguish between regions that are mathematically available and regions that are actually populated by observations. Density therefore enters the architecture as the empirical availability field against which grammatical pressures will later be computed.

The implementation is not intended as a broad robustness study, but the candidate-space construction was checked to ensure that it is not merely an artifact of a single accidental fit. Under ordinary perturbations of the focal token cloud, the same architecture continued to yield finite, convergent fitted densities. For present purposes, the relevant point is narrow: the empirical candidate space is numerically stable enough to support the formal comparison developed in the following sections.

That mediating role is central to the proposal. The same continuous constraints can interact differently with differently populated datasets without themselves being redefined. Density makes violations corpus-conditioned while leaving the grammatical side of the model separate. This is also why the present use of density should not be conflated with a token-cloud theory of category representation. The empirical density is introduced here not as a theory of what a category is, but as the continuous field within which candidates are available in greater or lesser concentration. The grammar enters separately through the constraint functions, and violation is the object produced by their interaction.

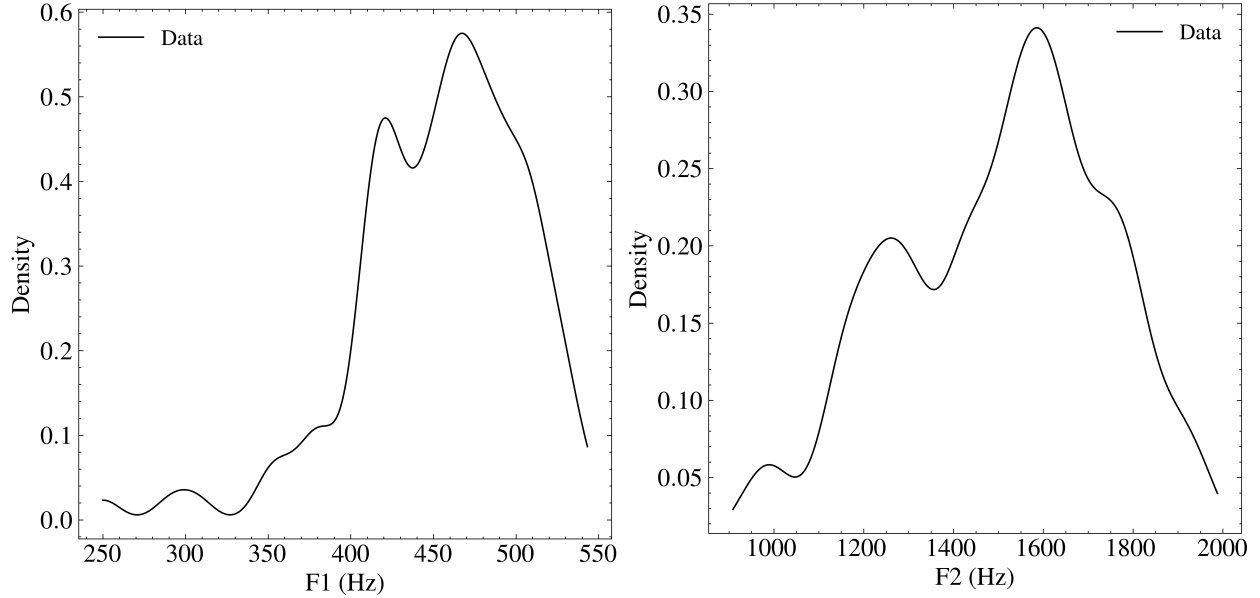


Figure 2: Estimated probability density of the present dataset. The density is estimated jointly over the two-formant space and plotted separately for F1 and F2 for visual clarity, with the axes returned to the original acoustic scale after estimation. This density supplies the empirical availability field whose local concentration is later interpreted grammatically through the violation link.

The next step is therefore not further description of the corpus, but specification of the grammatical pressures that will operate over this continuous empirical field. The following section introduces the continuous constraints whose local values will later be linked to density to produce explicit violations.

4 Continuous Constraints

Once the candidate space has been represented as a continuous empirical density, the next step is to specify the grammatical pressures that operate over it. In a discrete tableau, constraints can be applied to a finite list of candidates whose violation profiles are already available. In a continuous-candidate framework, that is no longer possible. Constraints must be stated as explicit functions over phonetic space. Their role in the present architecture is straightforward: they define where grammatical pressure is stronger or weaker before the next section asks how those local values become violations.

Continuous constraints are not themselves the novelty of the paper. Work at the phonetics–phonology interface has already shown that continuous phonetic spaces can be evaluated by explicit functions, including Flemming (2001) and Lefkowitz (2017). The narrower claim pursued here is that, once candidate availability is represented by an empirical density, those continuous constraint values still do not yet constitute violations. This section therefore introduces the constraint system that will later be linked to density. It is deliberately small: the argument only requires enough phonological structure to show that raw continuous pressure is not yet a usable violation object.

The implementation uses two constraints, one perceptual and one articulatory. That split keeps separate two familiar sources of phonological pressure: pressure to remain within a perceptually coherent category region, and pressure against costly production. In Boersma (1998), perceptual and articulatory evaluation are likewise distinct but related domains. The present paper does not attempt a full bidirectional grammar. It uses the smallest perceptual–articulatory inventory sufficient to make the link problem visible in a continuous candidate space.

The first constraint is perceptual. It models the idea that a phonological category favors a bounded acoustic region and penalizes departures from that region. A single logistic transition is not enough for that purpose, because it only captures movement into or out of a category on one side. What is required here is a bounded low-penalty interior with penalties that rise on both sides. The implementation therefore uses a paired logistic function over F1:

$$C_P(F1) = L - \frac{L^2}{(1 + e^{k_1(F1-\theta_1)})(1 + e^{-k_2(F1-\theta_2)})}. \quad (1)$$

This formulation creates a low-violation window inside a continuous F1 space while allowing penalties to rise outside that window. The parameters θ_1 and θ_2 define the upper and lower flanks of the window, and k_1 and k_2 determine how sharply those flanks rise. The point is not to impose a categorical cutoff, but to represent boundedness continuously. Because the two slope parameters need not match, the window can also be asymmetric. In the present /e/ implementation, that asymmetry provides a simple way of encoding the idea that movement toward neighboring regions

need not be equally costly on both sides of the category.

Table 4: Current implementation parameterization of the continuous constraints.

Constraint	Parameter	Value
Perceptual	L	1
Perceptual	θ_1	600
Perceptual	θ_2	345
Perceptual	k_1	1
Perceptual	k_2	7
Articulatory	$(TF1, TF2)$	(421, 1887)
Articulatory	$(NF1, NF2)$	(610, 1900)

Table 4 lists the working values used in the present implementation. Their function is to make the geometry of the constraint explicit rather than to provide final psychophysical estimates. In the current dataset, the lower flank is the one most visibly sampled by the attested F1 range, while the upper flank lies above the observed ceiling; this preserves a bounded perceptual window without making strong claims about exact boundary placement. For reproducibility, the table reports values in the original acoustic scale, but the implementation applies z-score standardization before computation so that the constraint functions and the empirical density are evaluated in the same coordinate system throughout.

The second constraint is articulatory. Whereas the perceptual constraint models category-sensitive acoustic well-formedness, the articulatory constraint models production-side cost. The relevant intuition is familiar from phonetically informed phonology: some realizations are costlier because they require greater displacement or effort (Boersma, 1998; Flemming, 1995; Flemming, 2004; Madrugá, 2017). The present implementation keeps that pressure simple. It does not attempt a full coarticulatory or biomechanical account, since the dataset does not contain the measurements needed for that. Instead, it defines articulatory cost relative to a target point and a neutral reference point.

The articulatory constraint is stated over the joint F1–F2 space:

$$C_A(F1, F2) = \sqrt{(F1 - TF1)^2 + (F2 - TF2)^2} \cdot \frac{\sqrt{(F1 - NF1)^2 + (F2 - NF2)^2}}{\sqrt{(TF1 - NF1)^2 + (TF2 - NF2)^2}}. \quad (2)$$

Here $(TF1, TF2)$ is the target configuration and $(NF1, NF2)$ is the neutral reference point. In the present parameterization, the target point (421, 1887) approximates the intended category location used in the analysis, while the neutral point (610, 1900) supplies a baseline relative to which displacement is scaled. The first factor in Eq. 2 measures distance from the target. The second scales that distance by realized displacement from the neutral point, normalized by the neutral-to-target displacement. The implemented articulatory pressure is therefore not just Euclidean distance in the F1–F2 plane; it is target deviation weighted by relative effort. Under this definition, realizations closer to the neutral baseline attenuate the target-distance penalty, whereas realizations requiring greater displacement amplify it.

As with the perceptual constraint, these values should be read as implementation parameters rather than as a complete articulatory theory of the vowel system. Their role is narrower and methodological: to introduce a production-side pressure over the same continuous domain as the perceptual constraint, so that the later link function has to relate more than one source of grammatical pressure to the empirical candidate field. In the computational implementation, a small positive constant ε is added to the Euclidean distances in the effort ratio for numerical stability; because the target and neutral points are distinct, this has no substantive effect on the current definition.

Taken together, the two constraints provide the minimal grammar required for the next stage of the argument. The perceptual constraint favors a bounded region of acoustic well-formedness; the articulatory constraint penalizes costly displacement in the same space. What they do not yet provide is a violation object. Their outputs are continuous values over the domain, but once candidate availability is represented by an empirical density, those values must still be transformed into violations that weighted evaluation can use.

That is the task of the next section. The comparison there holds the present constraint

functions fixed and asks how continuous constraint values should be linked to empirical density so as to yield an explicit, interpretable, and usable notion of violation.

5 Choosing a link function

At this point, the framework has continuous constraints and an empirical candidate density, but still lacks a violation object. A constraint specifies where phonological pressure is stronger or weaker; the empirical density $\hat{f}(x)$ specifies where candidate mass is concentrated. Neither quantity is itself a violation. In a discrete tableau, violations can be assigned directly to a listed candidate set. In a continuous space, that shortcut is unavailable. If local violation were simply identified with the raw output of a constraint, the model would evaluate the domain as though candidate availability were uniform, including unattested regions. Violations must therefore be defined through an explicit interaction between constraint value and empirical density.

The task of this section is to determine what kind of interaction yields a usable local violation magnitude. The question is not which algebraic transform is universally correct in the abstract. It is which transparent link produces a violation object that can support continuous weighted evaluation under the present architecture.

5.1 Desiderata for admissible link functions

An admissible link function should satisfy five conditions:

1. it should remain non-negative as a violation quantity;
2. it should attenuate violations in empirically dense regions;
3. it should remain stable in low-density or extrapolative regions;
4. it should preserve interpretability as a magnitude of violation;
5. it should remain compatible with later weighted harmonic and MaxEnt-style evaluation.

These desiderata are modest but sufficient for the present purpose. They do not prove that one formula is uniquely correct under every imaginable normalization. They instead provide an admissibility screen over a small family of transparent candidate links.

A first possibility is multiplication:

$$g_i^\times(x) = C_i(x)\hat{f}(x). \quad (3)$$

Multiplication lets density modulate the constraint directly, but it overly compresses the resulting surface. Wherever either factor is small, the product is pulled toward zero. Sparse regions are therefore not merely attenuated; they are largely flattened. That leaves too little structure for later weighted evaluation. Multiplication fails because it suppresses the very contrast a continuous violation surface must preserve.

A second possibility is subtraction:

$$g_i^-(x) = C_i(x) - \hat{f}(x). \quad (4)$$

Subtraction captures an important part of the desired logic. In denser regions, violation decreases; in sparser regions, the constraint remains more visible. But subtraction has two problems. First, it can become negative, which is awkward if violations are to remain magnitudes that feed weighted disharmony. Second, KDE values are often numerically small, so subtraction may modulate the constraint too weakly when the two components occupy different scales. The attenuation direction is right, but the resulting quantity is not reliably usable as a violation measure.

A third possibility is division:

$$g_i^\div(x) = \frac{C_i(x)}{\hat{f}(x)}. \quad (5)$$

Division sharpens the same attenuation logic. In denser regions the denominator increases and the resulting value decreases; in sparser regions it increases. The attraction is therefore real. The

problem is stability. Where density is very small, the ratio can explode; where density reaches zero, the expression becomes undefined. One could regularize the denominator with an ε term or impose a density floor, but then the link would depend on an auxiliary repair layer whose grammatical status is unclear. Direct division therefore fails the stability desideratum exactly where a continuous model most needs robustness.

The adopted relation preserves what is useful in subtraction and division while avoiding their strongest defects. Subtraction contributes the correct attenuation direction; division contributes sharper contrast between dense and sparse regions. Exponentialization retains that logic without literal negativity and without literal division by zero. Begin with a ratio between powers of the same base:

$$v_i(x) = \frac{y^{C_i(x)}}{y^{\hat{f}(x)}}. \quad (6)$$

Choosing base e gives

$$v_i(x) = \frac{e^{C_i(x)}}{e^{\hat{f}(x)}}, \quad (7)$$

which simplifies to

$$v_i(x) = e^{C_i(x) - \hat{f}(x)}. \quad (8)$$

This derivation matters because it makes clear what the adopted link is doing. The exponent preserves the subtractive relation between constraint value and empirical density, while the exponential transform guarantees strict positivity and strengthens differences that would otherwise remain numerically weak. Because the denominator is no longer retained as a literal divisor, the pathology of direct division at $\hat{f}(x) = 0$ is avoided. The result is a local violation magnitude that stays positive, responds to empirical concentration, and remains computationally usable for weighted evaluation.

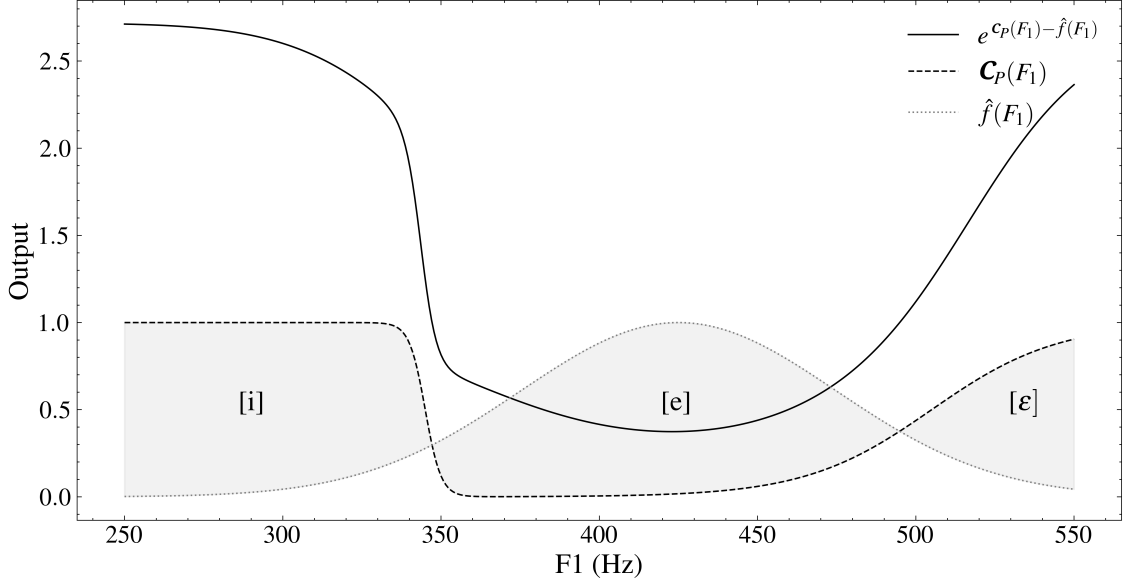


Figure 3: Exponentialized link, $v_i(x) = e^{C_i(x) - \hat{f}(x)}$. The resulting surface attenuates violation in dense regions while remaining strictly positive and computationally usable.

The choice of base e is pragmatic rather than theoretical. The natural exponential maps additive differences in the exponent onto multiplicative differences in the output, so even modest local differences between constraint value and density become consequential without introducing a separate amplification parameter. The present implementation therefore fixes the exponentialized relation as the direct link between continuous grammatical pressure and empirical candidate availability.

This also clarifies the issue of commensurability. The framework does not assume that raw constraint outputs and raw density values are pretheoretically the same kind of quantity. The need for a link function arises precisely because they are not. The link is the transformation that makes them jointly usable inside a single evaluative object. In that sense, commensurability is not a prior assumption of the model; it is the result of the transformation used to define violation.

The comparison among candidate links should be understood in that light. Multiplication is rejected because it over-compresses the surface. Subtraction is rejected because it does not guarantee positivity. Direct division is rejected because it is unstable in sparse regions. The exponentialized relation is adopted not because it wins every possible predictive comparison, but because among the

transparent direct links considered here it best satisfies the constitutive requirements of a usable violation object under the present architecture.

The point of the comparison is architectural before it is predictive. Multiplication, subtraction, and direct division are not rejected because they fail to outperform a preferred competitor by some small empirical margin. They are rejected because each fails, in a different way, a constitutive requirement of a usable violation object under the present desiderata: multiplication over-compresses the evaluative surface, subtraction sacrifices guaranteed positivity, and direct division becomes fragile in sparse regions.

This matters because goodness of fit alone does not settle the choice of link. A transformation may in principle approximate an empirical distribution relatively well while still being unsatisfactory as a violation function if it permits negative penalties, erases local contrast through compression, or depends on unstable behavior precisely where empirical density is low. Conversely, a link may be preferred even if some rival transformation fits a particular dataset better, provided that the rival does so only by failing one of those constitutive requirements or by relying on auxiliary repair devices.

That is the sense in which the exponentialized relation is adopted here. The claim is not that it is the universal winner of every predictive comparison, nor that no alternative rescaled formulation could ever be constructed. The narrower claim is that, among the transparent direct links considered in this article, it is the strongest admissible direct link under the stated desiderata and the present implementation: it preserves positivity, yields the intended attenuation in dense regions, avoids the singularity problem of direct division, and remains usable for downstream weighted evaluation without additional rescue calibration.

Table 5: Conceptual comparison of transparent direct links between continuous constraint value and empirical density, including attenuation in dense regions and stability in sparse regions.

Link	Positivity	Attenuation	Stability	Main defect
Multiplication	Yes	Over-compressed	Yes	Collapses local contrast
Subtraction	No	Yes	Yes	Permits negative violations
Direct division	Yes	Yes	No	Low-density instability
Exponentialized	Yes	Yes	Yes	Adopted direct link

The conclusion of this comparison is therefore deliberately narrow. The exponentialized relation is adopted here not because it guarantees the best fit under every conceivable metric, but because it is the most satisfactory *direct* link under the present theoretical requirements. It preserves positivity, supports the intended attenuation of violations in empirically dense regions, avoids the singular behavior of direct division, and remains usable in downstream weighted evaluation without auxiliary repair devices. In that sense, it provides the clearest admissible basis for the continuous violation theory developed in the sections that follow.

6 Continuous violations

The previous section selected the exponentialized relation as the admissible link between continuous constraint value and empirical density. The next step is to define what a violation is once candidates are no longer treated as pre-discretized tableau rows. In the present framework, violations are not inherited from bounded candidates and are not identified with raw constraint values alone. They are defined pointwise over a continuous domain and accumulated over selected intervals or regions by integration. The formal task of this section is therefore straightforward: to state the violation object that follows from the link already adopted.

Let $C_i(x)$ be the value returned by constraint i at point x , and let $\hat{f}(x)$ be the empirical

density evaluated at that same point. The pointwise violation quantity is

$$v_i(x) = e^{C_i(x) - \hat{f}(x)}. \quad (9)$$

This is the local object on which the rest of the continuous grammar depends. Once it is defined, interval- and region-level quantities follow by accumulation rather than by assigning a uniform penalty to a pre-delimited candidate.

For a one-dimensional interval I , the violation associated with constraint i is

$$V_i(I) = \int_I v_i(x) dx = \int_I e^{C_i(x) - \hat{f}(x)} dx. \quad (10)$$

For a two-dimensional region Ω , the same logic yields

$$V_i(\Omega) = \int_{\Omega} v_i(x) dx = \int_{\Omega} e^{C_i(x) - \hat{f}(x)} dx. \quad (11)$$

These definitions make explicit what a continuous violation is in the present architecture: a region's violation burden is the accumulated contribution of its local points, where each local contribution is determined jointly by constraint value and empirical concentration.

When reporting domains differ in size, raw integrated mass is not always the most comparable quantity. It is therefore useful to define mean local violation,

$$\bar{V}_i(\Omega) = \frac{1}{|\Omega|} \int_{\Omega} v_i(x) dx, \quad (12)$$

with the one-dimensional analogue $\bar{V}_i(I) = |I|^{-1} \int_I v_i(x) dx$. Here $V_i(\cdot)$ measures total accumulated burden, while $\bar{V}_i(\cdot)$ measures average local severity independently of window size. When useful, either quantity may also be reported as a share of the total violation mass in the analyzed domain. The important point is that all such summaries are read off from the same continuous pointwise field. The grammar therefore evaluates the space locally first and only then permits broader or narrower

summaries of the resulting surface.

Two cases make the definitions concrete. The first is the perceptual constraint, which in the present implementation depends only on F1. Because the link operates between a constraint and an empirical density of matching dimensionality, the relevant empirical object is not the full joint density over F1 and F2, but the F1 marginal derived from it. If the empirical candidate space is estimated as $\hat{f}(F1, F2)$, the corresponding one-dimensional density is

$$\hat{f}_{F1}(F1) = \int_{a_{F2}}^{b_{F2}} \hat{f}(F1, F2) dF2. \quad (13)$$

In the numerical implementation, this marginal is computed over the same bounded F2 domain used in the analysis and then renormalized over F1. The continuous perceptual violation associated with an interval $I \subseteq [a_{F1}, b_{F1}]$ is therefore

$$V_P(I) = \int_I e^{C_P(F1) - \hat{f}_{F1}(F1)} dF1. \quad (14)$$

The perceptual case is therefore one-dimensional not because the data cease to be joint, but because the constraint itself is one-dimensional and the empirical density must be matched to that constraint accordingly. Figure 4 illustrates this case.

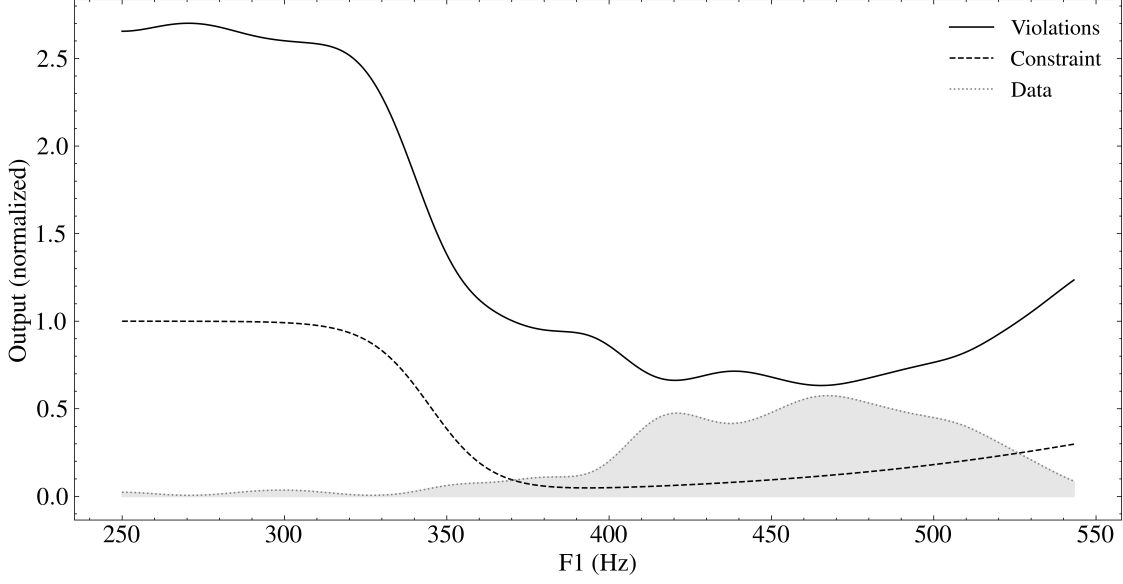


Figure 4: Continuous perceptual violations over an F1 interval.

The second case is the articulatory constraint, which is defined jointly over F1 and F2. Here no marginalization is required, because both the constraint and the empirical candidate space are two-dimensional. The relevant object is therefore the full joint density itself, and the corresponding integrated articulatory violation over a region Ω is

$$V_A(\Omega) = \iint_{\Omega} e^{C_A(F1, F2) - \hat{f}(F1, F2)} dF1 dF2. \quad (15)$$

This contrast with the perceptual case is methodological rather than theoretical in any deeper sense. It follows from dimensional matching: where the constraint is one-dimensional, the empirical density must be marginalized to one dimension; where the constraint is two-dimensional, the full joint density enters directly. Figure 5 shows one-dimensional projections of this two-dimensional articulatory surface onto the F1 and F2 axes.

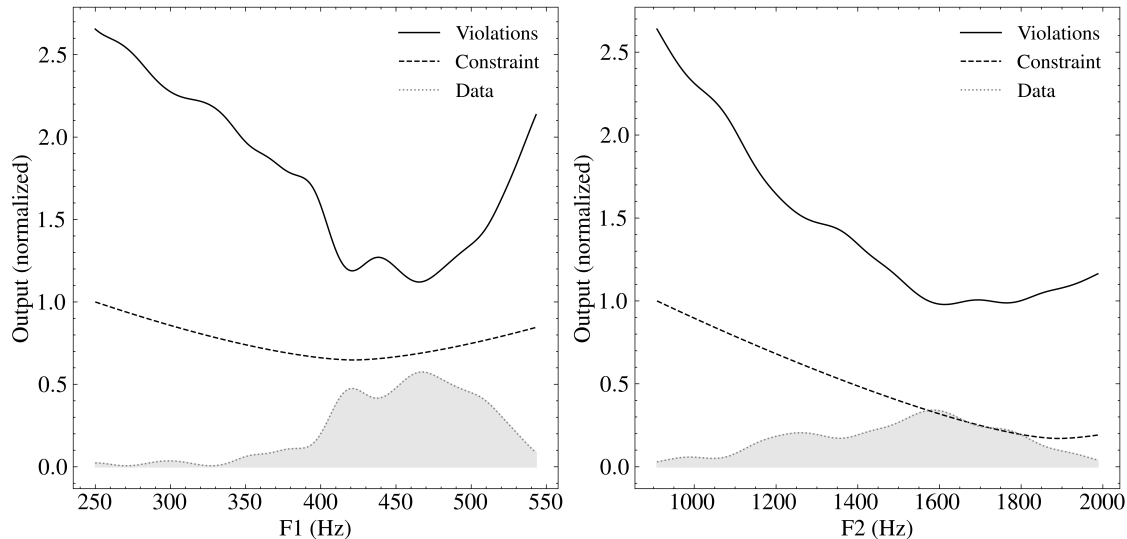


Figure 5: One-dimensional projections of continuous articulatory violations from the joint F1–F2 space.

These definitions are the point at which the paper’s central proposal becomes operational. Constraint values are no longer treated as if they already were violations, and candidate availability is no longer supplied by a pre-binned inventory. Instead, the violation object is defined explicitly as the local interaction between grammatical pressure and empirical density, and integrated values summarize that interaction over selected domains. Once this object is in place, the rest of weighted phonological evaluation can proceed in familiar form. The next section shows how these continuous violations feed Harmony and a local stability diagnostic over the same continuous candidate space.

7 Harmony and stability

Once violations are defined pointwise over a continuous candidate space, the next question is what ordinary weighted evaluation looks like downstream. The answer is deliberately conservative. The present paper does not replace Harmonic Grammar with a new evaluative mechanism; it extends familiar weighted evaluation to a setting in which the violation object has already been defined over a continuous field. This continuity also makes available one additional diagnostic of the resulting landscape. Harmony captures accumulated weighted penalty. Stability, treated here as a secondary

diagnostic, summarizes how that harmonic field changes across a selected domain. The main innovation of the paper remains the definition of continuous violation; harmony and stability are downstream consequences of that definition.

The first consequence concerns harmonic evaluation itself. In Harmonic Grammar, harmony is a weighted sum of violations, with better candidates corresponding to lower weighted totals (Flemming, 2021; Goldwater and Johnson, 2003; B. Hayes and Wilson, 2008). Once violations have been defined over a continuous phonetic space, the same logic carries over directly. Let I be a reporting interval and let $V_i(I)$ be the integrated violation associated with constraint i . A continuous analogue of harmonic evaluation is:

$$H(I) = \sum_{i=1}^n \lambda_i V_i(I), \quad (16)$$

where λ_i is the weight assigned to constraint i . Nothing special happens at this stage. Once a continuous violation quantity exists, harmony follows by ordinary weighted summation. The theoretical issue was upstream: what kind of violation object is being weighted once candidates are continuous.

In the present two-constraint implementation, harmony is the weighted sum of a perceptual and an articulatory term:

$$H(I) = \lambda_P V_P(I) + \lambda_A V_A(I), \quad (17)$$

and, at the pointwise level,

$$H(F1, F2) = \lambda_P v_P(F1) + \lambda_A v_A(F1, F2). \quad (18)$$

This distinction matters. Integrated quantities support comparison over selected intervals or regions, while the pointwise definition yields a continuous evaluative field over the candidate space itself. Harmony therefore remains a weighted grammatical score, not yet a probability. Its role is to summarize how severely a region is penalized once continuous constraint pressure has been linked

to empirical density.

Once harmony is defined pointwise, one may also ask how that weighted field changes across a selected domain. Under the present implementation, this question is tracked by a derived diagnostic called *stability*. Stability is not a second kind of violation, not a second notion of well-formedness, and not a central claim of the article. It is an implementation-level summary of local change in the harmonic field, included here to illustrate one kind of information that becomes available once evaluation is defined continuously rather than over pre-binned candidates. Harmony asks how much weighted penalty a region accumulates. Stability, in the narrower sense adopted here, asks whether that field is locally flatter or steeper on average across a reporting window.

Formally, if harmony is defined pointwise as $H(x)$ over a continuous domain, then local change is described by its gradient. In the present two-dimensional case, the relevant partial derivatives are with respect to F1 and F2. The stability score S over a region Ω is defined as the size-normalized integral of their summed contribution:

$$S(\Omega) = \frac{1}{|\Omega|} \iint_{\Omega} \left(\frac{\partial H}{\partial F1} + \frac{\partial H}{\partial F2} \right) dF1 dF2, \quad (19)$$

where $|\Omega|$ is the area of the reporting domain. The one-dimensional analogue for an interval I is

$$S(I) = \frac{1}{|I|} \int_I \frac{dH}{dx} dx. \quad (20)$$

Because this quantity is normalized by domain size, it is not a raw accumulation of change but an average change measure over the selected window. Values near zero indicate a relatively flat harmonic surface; larger absolute values indicate steeper average change. Under the present numerical implementation, the sign indicates whether harmony tends on balance to increase or decrease over the chosen domain given the direction of integration. This quantity should be interpreted narrowly. It is not intended as a coordinate-invariant or theory-general definition of stability, but as one convenient summary of average local slope under the present implementation. Because it is computed from directional derivatives over the chosen coordinates and reporting

domain, its magnitude and sign depend on those analytic choices. For that reason, the diagnostic is used here illustratively rather than foundationally: its role is to show that a continuous grammar preserves inspectable internal geometry that pre-evaluative binning would flatten, not to propose a universally privileged measure of landscape structure.

The usefulness of this diagnostic is correspondingly limited but still informative. Harmony and stability do not answer the same question. Harmony measures weighted violation burden over a selected domain. Stability, in the present implementation, gives a compact indication of whether that burden is relatively even or changing more sharply across the same window. A region may therefore have relatively low harmonic burden while still exhibiting appreciable internal change, or be comparatively flat while remaining uniformly poor. Read in that limited way, the diagnostic helps distinguish low-penalty plateaus from steeper portions of the same continuous field.

This is one practical consequence of continuous evaluation. In a discretized analysis, a candidate row may receive a weighted total, but the row itself does not preserve internal geometry. Once all values inside a bin inherit one violation profile, the grammar cannot directly register whether the harmonic field inside that bin is flat, steep, or internally asymmetric. In the present setting, that information is preserved because harmony is defined pointwise before any regional summary is taken. Stability is simply one way of reading that preserved structure.

The framework therefore supports one primary and one secondary downstream reading of continuous candidate structure. Weighted harmony preserves the familiar Harmonic-Grammar logic of accumulated penalty. Stability adds only an implementation-specific slope diagnostic over that same field. Even in that modest role, however, it makes visible a kind of internal organization that discretized candidates obscure: once evaluation is defined pointwise, a reporting window can be characterized not only by how penalized it is overall, but also by whether the surrounding harmonic surface is comparatively flat or sharply changing. The next section turns from that descriptive diagnostic to the probabilistic side of the architecture and asks how the continuous weighted field is converted into a MaxEnt-style output density.

8 Optimization, MaxEnt and empirical fit

Once continuous violations have been defined, the remaining question is what distribution a weighted grammar predicts over the candidate space and how closely that distribution approximates the observed concentration structure. The present framework retains the downstream logic of discrete MaxEnt phonology (Goldwater and Johnson, 2003; B. Hayes and Wilson, 2008), but applies it to a continuous candidate space rather than to a finite candidate list. The unnormalized MaxEnt field is therefore

$$\tilde{f}_{\text{MaxEnt}}(x) = e^{-1-\lambda_0-\sum_{i=1}^n \lambda_i v_i(x)}, \quad (21)$$

where λ_0 is an intercept term and the $v_i(x)$ are the linked, non-negative violations introduced above. In the present two-dimensional case,

$$\tilde{f}_{\text{MaxEnt}}(F1, F2) = e^{-1-\lambda_0-\lambda_P v_P(F1)-\lambda_A v_A(F1, F2)}. \quad (22)$$

The normalized output density is then

$$f_{\text{MaxEnt}}(F1, F2) = \frac{\tilde{f}_{\text{MaxEnt}}(F1, F2)}{\iint_{\mathcal{D}} \tilde{f}_{\text{MaxEnt}}(u, v) du dv}, \quad (23)$$

where $\mathcal{D}$ is the evaluation domain. In the current implementation, this normalization is carried out numerically over the analysis grid before model–data comparison. The resulting object is therefore a proper output density over the continuous candidate space rather than an unscaled penalty surface. Because the normalization in (23) divides by the integral of the same unnormalized field, λ_0 functions only as an implementation-level offset on $\tilde{f}_{\text{MaxEnt}}$ and cancels in the normalized density; it is reported below solely for transparency and reproducibility, not as an independently interpretable grammatical weight.

The fitted weights are chosen by minimizing Kullback–Leibler divergence between the normalized empirical density and the normalized modeled density (Kullback and Leibler, 1951;

Cover and Thomas, 2006):

$$D_{\text{KL}}(\hat{f} \parallel f_{\text{MaxEnt}}) = \int \hat{f}(x) \log \frac{\hat{f}(x)}{f_{\text{MaxEnt}}(x)} dx. \quad (24)$$

KL minimization is appropriate here because the empirical object and the grammar’s output are both continuous densities over the same space. The question is whether the weighted grammar reconstructs the broad structure of the empirical candidate space better after fitting than under an untrained baseline.

For the present implementation, optimization yields the values in Table 6. Read as an internal before-and-after comparison, the reduction from 4.233 to 0.351 shows that the fitted grammar approximates the empirical density substantially better than the untrained starting weights do. The optimizer itself is only a practical choice; a bounded L-BFGS-B routine is used because the objective is smooth and nonlinear.

Quantity	Value
λ_0 (implementation offset)	1.027
λ_P	0.417
λ_A	0.018
Pre-optimization $D_{\text{KL}} (\lambda_P = \lambda_A = 1)$	4.233
Post-optimization D_{KL}	0.351

Table 6: Optimization summary for the present implementation.

The fitted unnormalized field is therefore

$$\tilde{f}_{\text{MaxEnt}}(F1, F2) = e^{-2.027 - 0.417 v_P(F1) - 0.018 v_A(F1, F2)}. \quad (25)$$

After normalization, the shape of the fitted density is determined primarily by the perceptual term, with a smaller articulatory contribution. Under the present implementation, the grammar is therefore

perceptually dominated but articulatorily conditioned.

Figure 6 plots the normalized marginal output predicted by the fitted grammar against the normalized empirical density along each formant dimension. The figure should be read as the output of the weighted grammar after continuous violations have been exponentiated and normalized, not as a direct replotting of the token cloud.

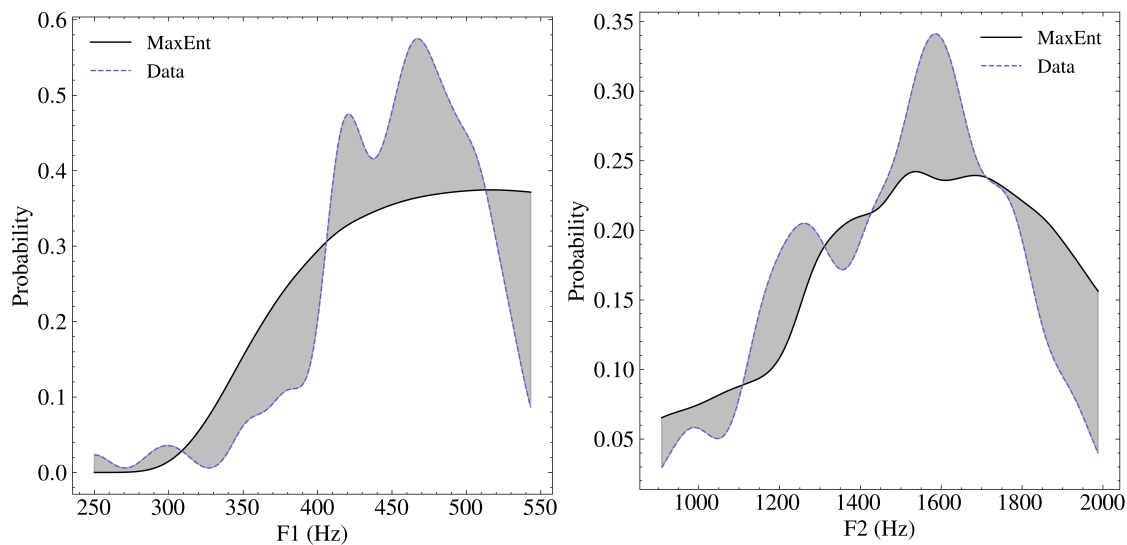


Figure 6: Normalized marginal comparison between the fitted MaxEnt output density and the empirical density for the present /e/ space.

The fitting result is used here in a narrow way. It does not establish final empirical adequacy. Its role is to show that, once continuous violations have been defined, ordinary weighted phonological evaluation can be carried through to a continuous output density that improves substantially on an untrained baseline. The next section turns to the phonological consequences of that fitted grammar for selected regions of the present /e/ space.

9 Phonological consequences for the present /e/ space

The previous sections established the continuous violation object, showed how it feeds Harmony and a secondary local-slope diagnostic, and fitted a MaxEnt grammar over the present /e/ space. The remaining question is narrower: what phonological interpretation does that fitted

grammar support in this implementation? The answer is not that it recovers hidden subcategories. Rather, it allows selected windows within one continuous category to be used as diagnostic probes. These windows are illustrative reporting domains, chosen to span the attested space coarsely rather than discovered by the model. They are analytic slices of the same candidate field, queried to show whether different neighborhoods of the category receive different profiles of penalty, local change, and output support.

Three quantities are relevant here. Harmony summarizes weighted penalty over a selected domain. Stability, used only as a secondary diagnostic, summarizes average local change in the Harmony field over that domain under the present implementation. Output support is read from the fitted MaxEnt density, both jointly over F1–F2 and marginally along each dimension. Together, these quantities allow the grammar to say more than that /e/ occupies a broad acoustic region: they make it possible to ask whether some parts of that region are less supported, more heavily penalized, or associated with flatter versus more steeply changing portions of the same continuous field.

Table 7 gives one broad-region comparison. The windows are reporting domains within the same continuous candidate space, not discrete outputs and not psychological subcategories. Even at this coarse scale, the fitted grammar does not treat the /e/ space as phonologically flat. The lower region has much higher Harmony and far lower joint output probability than either the middle or upper region. The middle and upper regions are both substantially better supported, though not in identical ways: the middle region has slightly higher Harmony than the upper region but somewhat higher joint probability. That pattern matters because it shows that support in the fitted grammar is not exhausted by one scalar penalty value alone. It reflects the distribution of the weighted field over the continuous space.

Region ($F1$, $F2$ in Hz)	H	S	$P_{F1,F2}$	P_{F1}	P_{F2}
Lower region: 250–350, 908–1300	11.859	−8.831	0.008	0.057	0.205
Middle region: 350–450, 1300–1700	0.983	−0.612	0.204	0.421	0.476
Upper region: 450–543, 1700–1987	0.574	+0.138	0.163	0.517	0.314

Table 7: Broad-region illustrative diagnostic windows within the present continuous /e/ space. H is Harmony, S is Stability, $P_{F1,F2}$ is the joint output probability, and P_{F1} and P_{F2} are marginal output probabilities.

Read together, Harmony and Stability make the contrast more precise, but in different ways. Lower Harmony corresponds to regions the fitted grammar treats as better formed overall. Stability does not add a second notion of well-formedness; in the present implementation, it serves only as a compact summary of average local change in the Harmony field over the selected window. On that narrower reading, the lower region is not only weakly supported but also associated with a more sharply changing portion of the weighted field. The middle region lies in a less strongly penalized and less sharply changing portion of the same space, while the upper region combines the lowest Harmony with Stability closer to zero. The point is therefore not simply that one region is “better” than another. It is that continuous evaluation makes it possible to distinguish differently penalized and differently shaped neighborhoods within the internal evaluative geography of the category.

The marginal probabilities reinforce that conclusion. In all three windows, separate support along $F1$ and $F2$ is higher than the corresponding joint support. That is expected mathematically, but it is also phonologically informative in the present architecture. Perceptual pressure is evaluated over the one-dimensional $F1$ projection, whereas articulatory pressure is evaluated over the joint $F1$ – $F2$ space. The divergence among P_{F1} , P_{F2} , and $P_{F1,F2}$ therefore shows that internal support is distributed differently across dimensions. A region may be relatively well supported on one axis while remaining less favored jointly. Continuous evaluation thus makes it possible to represent the internal geography of the category without replacing that category with pre-evaluative bins.

A single overlap comparison is enough to make the same point at a finer scale. Table 8 compares two intersecting windows taken from the same continuous field. Because they overlap

broader regions rather than partitioning the space exhaustively, they show that the fitted grammar can be queried at different scales without changing what counts as a candidate.

Overlap region ($F1$, $F2$ in Hz)	H	S	$P_{F1,F2}$
300–400, 1200–1500	2.184	−1.219	0.071
400–500, 1500–1900	0.578	−0.129	0.251

Table 8: Overlap-region illustrative diagnostic windows within the same continuous /e/ space. These are intersecting reporting windows queried against one continuous grammar.

The contrast between these overlap probes is internally consistent with the broader pattern. The lower-overlap window has higher Harmony, more negative Stability, and lower joint probability. The higher-overlap window has lower Harmony, Stability closer to zero, and much greater joint support. Under the present implementation, this is consistent with the second window lying in a more favorable and less sharply changing portion of the weighted field. The overlap comparison therefore adds one limited but useful point: the grammar does not merely distinguish broad regions, but can also differentiate nearby intersecting reporting domains within the same continuous category.

This is the level at which the present phonological claim should be understood. The analysis is not intended to discover hidden categorical units inside /e/. It shows, more narrowly, that once violations are defined over a continuous candidate space, the fitted grammar can return region-sensitive phonological interpretation over attested vowel data. In the present implementation, some neighborhoods of the category are weakly supported and more heavily penalized, while others are better supported and associated with flatter or less sharply changing portions of the weighted field. That is the payoff of the continuous architecture here: it lets ordinary weighted phonological evaluation operate over the internal structure of one continuous vowel category rather than over a discretized surrogate.

10 Scope and limitations

The contribution of this article is methodological rather than population-wide. It introduces an explicit violation object for continuous-candidate evaluation and shows that this object can support downstream weighted computation over attested acoustic data. The scope conditions below therefore delimit the level of empirical generality claimed here without changing the paper’s central result.

10.1 Corpus scope

The present implementation uses 114 tokens from three adult male speakers and a single vowel class represented in a two-formant space. That empirical base is intentionally narrow. The relevant question, however, is not whether the current sample exhausts the empirical range of continuous phonological phenomena, but whether the proposed architecture behaves coherently over a real continuous candidate space. In that narrower sense, the current implementation is sufficient: under ordinary perturbation of the focal dataset, the fitted grammar remains convergent and finite rather than collapsing numerically.

The same point applies to the article’s focus on F1 and F2. A familiar two-dimensional vowel space makes the formal architecture inspectable without unnecessary dimensional complexity. Nothing in the proposal depends on formants as the only relevant continuous variables. The same logic could in principle be extended to duration, pitch, intensity, timing, or higher-dimensional combinations, provided that continuous constraints and an empirical candidate density can be defined over those spaces. The present corpus therefore fixes a tractable implementation domain; it does not define the outer limit of the framework.

For the same reason, the restricted constraint inventory should be read as a methodological simplification. One perceptual pressure and one articulatory pressure are enough to expose the link architecture transparently. The paper does not claim that these two constraints exhaust the grammar of the phenomenon.

10.2 Preprocessing scope

The preprocessing choices should be understood in the same way: as implementation conditions rather than as the substance of the claim. In the current dataset, Lobanov normalization was explored initially, but it produced numerical problems for the kernel density estimation used here. Z-score standardization was therefore adopted because it yielded a stable empirical surface for the present implementation. Different choices of scaling, bandwidth, or numerical estimation could reshape the estimated field, but they would not change the formal problem addressed by the paper: how continuous constraint values and empirical density are linked to define violations.

10.3 Parameter status

The current parameterization is likewise best read as an implementation of the architecture rather than as a final substantive estimate. The threshold and slope values used for the perceptual constraint are working values chosen to instantiate a bounded category window explicitly. The articulatory target and neutral point serve a parallel role: they provide a second independently interpretable pressure without claiming to supply a complete biomechanical or gestural model. A richer empirical program could replace these values with more fully grounded perceptual or articulatory estimates, but the formal point of the present article does not depend on that further step.

10.4 Scale, circularity, and model-conditionality

A separate issue concerns scale. Raw density values and raw constraint outputs need not begin as identical kinds of magnitude. The purpose of the link function is precisely to make them jointly usable in grammatical evaluation. On this view, commensurability is not something the model assumes in advance; it is something the violation transformation is introduced to secure.

A related issue concerns circularity. Empirical density enters the architecture twice: upstream as the conditioning field for local violation and downstream as the target for the fitted output density. Those roles are related but distinct. Upstream, density modulates how constraint-defined pressure is

realized locally across the candidate space. Downstream, the fitted grammar asks whether weighted evaluation built from those violations can reproduce the broad structure of the empirical field better than an untrained baseline. That dual role is therefore part of the model specification, not an unnoticed collapse of data into theory.

These limitations matter, but they do not alter the paper’s main claim. Even under implementation-conditional choices, the present grammar supports interpretable region-sensitive evaluation and a fitted continuous output density over the /e/ space. The conclusion returns to the broader consequence: once candidates remain continuous, constraint-based phonology requires an explicit violation theory, and with such a theory in place, ordinary weighted evaluation can operate directly over continuous empirical structure.

11 Conclusion

This paper has addressed a specific theoretical problem for constraint-based phonology: once candidates remain continuous, violation computation can no longer be inherited from a pre-delimited candidate set. A continuous candidate space therefore requires an explicit account of how grammatical pressure becomes local violation. The proposal developed here supplies that missing object by linking continuous constraint values to empirical density. In this architecture, density functions as an availability field over the candidate space, constraints remain the source of grammatical pressure, and violations are the quantities produced by their interaction. That is the paper’s central claim.

Within the family of transparent links examined here, the exponentialized relation emerges as the strongest admissible option under the present desiderata and implementation. Its advantage is architectural rather than merely numerical: it preserves non-negativity, remains sensitive to empirical concentration, avoids the instability of direct division in sparse regions and supports downstream weighted evaluation without additional repair machinery. The claim is intentionally narrow. It is not that every possible alternative has been ruled out in principle, but that the present proposal yields a

coherent and explicit definition of continuous violation where tableau-style inheritance is no longer available.

Once that object is defined, the rest of the evaluative pipeline can remain recognizably phonological. Continuous violations can be integrated over intervals or regions, combined in weighted harmonic evaluation and transformed into a continuous MaxEnt-style output density. The fitted implementation shows that this architecture can operate over attested acoustic data rather than over a discretized surrogate. In the present /e/ space, optimization yields fitted weights $\lambda_P = 0.417$ and $\lambda_A = 0.018$, and the Kullback–Leibler divergence decreases from 4.233 under the untrained starting weights to 0.351 after fitting. Under this implementation, the resulting grammar is perceptually dominated but articulatorily conditioned. The point of these results is not to claim a final empirical account of Brazilian Portuguese pretonic /e/, but to show that ordinary weighted phonological evaluation can operate coherently once violations are defined explicitly for continuous candidates.

The broader consequence is methodological. A continuous category need not be replaced by pre-evaluative bins in order to receive phonological interpretation. If violations are defined pointwise over the space, then category-internal gradient structure can remain inside grammatical evaluation rather than being flattened into candidate intervals in advance. The empirical implementation here is deliberately narrow, but it is sufficient to show that the framework supports continuous evaluation, weighted comparison and region-sensitive interpretation over a real vowel space.

Future work should test the framework under broader empirical and theoretical conditions. On the empirical side, larger corpora, additional speakers and other phonetic dimensions will show how robust the present architecture remains outside the current implementation domain. On the theoretical side, richer constraint inventories, alternative link functions and other sources of candidate structure remain open questions. What this paper establishes is the more basic point on which those extensions depend: constraint-based phonology lacks an explicit violation theory for continuous candidate spaces, and such a theory can be supplied in a form that preserves the logic of weighted phonological evaluation.

Conflict of Interest

The author declares that there are no financial, personal, academic, political, or commercial conflicts of interest that could have influenced the work reported in this article.

Ethics and Regulatory Note

The data used in this study were collected with approval within the Brazilian CEP/Conep system for the ethical review of research involving human participants, that is, through a Research Ethics Committee (CEP) and, where applicable within the national system, under the regulatory coordination of the National Research Ethics Commission (Conep). The Certificate of Presentation for Ethical Consideration (CAAE) number is 64773022.8.0000.8142, and the receipt number is 123890/2022.

Data Availability Statement

The data, materials, and analytical routines that support the results reported in this article are available in an open repository. The repository includes the corpus used in the analysis and the files necessary for its methodological reconstruction. Repository address: <https://github.com/alexandre-barroso/ContinuousConstraints/>. Persistent identifier (DOI): <https://doi.org/10.5281/zenodo.19411186>.

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